Homework 8

Grant Talbert
December 8, 2025

Boston University

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Problem 1.1. *Let M be a smooth manifold, $V \in \mathfrak{X}(M)$, and $\eta, \omega \in \Omega^*(M)$. Then*

$$\mathcal{L}_V(\eta \wedge \omega) = \mathcal{L}_V \eta \wedge \omega + \eta \wedge \mathcal{L}_V \omega.$$

Proof. Let θ_t be the flow of V. By definition, $\mathcal{L}_V \alpha = \frac{d}{dt}\big|_{t=0} \theta_t^* \alpha$. By Lee 14.16,

$$\theta_t^*(\eta \wedge \omega) = \theta_t^* \eta \wedge \theta_t^* \omega.$$

Applying the product rule for regular differentiation,

$$\begin{aligned}
\mathcal{L}_V(\eta \wedge \omega) &= \frac{d}{dt}\bigg|_{t=0} (\theta_t^* \eta \wedge \theta_t^* \omega) \\
&= \left(\frac{d}{dt}\bigg|_{t=0} \theta_t^* \eta \right) \wedge \theta_0^* \omega + \theta_0^* \eta \wedge \left(\frac{d}{dt}\bigg|_{t=0} \theta_t^* \omega \right) \\
&= (\mathcal{L}_V \eta) \wedge \omega + \eta \wedge (\mathcal{L}_V \omega),
\end{aligned}$$

where we used that $\theta_0 = \mathrm{id}$ so $\theta_0^* = \mathrm{id}$. ■

Problem 14.1

Problem 2.1. *Show that covectors $\omega^1, \ldots, \omega^k$ on a finite dimensional vector space are linearly dependent if and only if $\omega^1 \wedge \cdots \wedge \omega^k = 0$.*

Proof. ($\Longrightarrow$) Let the collection of covectors be linearly dependent. Then there is $1 \leq i \leq k$ and some collection of constants $\{c_j\}_{1 \leq j \leq k, j \neq i}$ such that

$$\omega^i = \sum_{1 \leq j \leq k, j \neq i} c_j \omega^j.$$

By multilinearity of the exterior product,

$$\omega^1 \wedge \cdots \wedge \left(\sum_{1 \leq j \leq k, j \neq i} c_j \omega^j \right) \wedge \cdots \wedge \omega^k = \sum_{1 \leq j \leq k, j \neq i} c_j \omega^1 \wedge \cdots \wedge \omega^j \wedge \cdots \wedge \omega^k.$$

The indicated ω^j appears in the ith component of the wedge product by hypothesis. However, another ω^j appears in the jth component of the wedge product by definition. By hypothesis, since $j \neq i$, there are two ω^js in the wedge product, and because the wedge product is alternating it follows that the wedge product is 0. Thus each term in the sum is 0.

($\Longleftarrow$) We prove the contrapositive. Let $\{\omega^i\}_{i=1}^k$ be linearly independent, and thus there exists a basis $\{\omega^i\}_{i=1}^n$ for V^* extending the linearly independent set. By Lee 14.8, there is a basis for $\Lambda^k(V^*)$ given by

$$\mathcal{B} \coloneqq \left\{ \omega^I \mid I \text{ is a } k\text{-multi-index} \right\} \subseteq \Lambda^k(V^*)$$

By Lee 14.10, $\omega^i \wedge \omega^j = \omega^{(i,j)}$. Thus there is a basis

$$\mathcal{B} = \left\{ \omega^{i_1} \wedge \cdots \wedge \omega^{i_k} \right\}_{1 \leq i_1 < \cdots < i_k \leq n} \subseteq \Lambda^k(M).$$

In particular,

$$\omega^1 \wedge \cdots \wedge \omega^k \in \mathcal{B}.$$

Since a basis contains all nonzero vectors, it follows that $\omega^1 \wedge \cdots \wedge \omega^k \neq 0$. ■

Problem 14.6

Problem 3.1. *Define the 2-form on $\mathbb{R}^3$ by*

$$\omega = x\mathrm{d}y \wedge \mathrm{d}z + y\mathrm{d}z \wedge \mathrm{d}x + z\mathrm{d}x \wedge \mathrm{d}y.$$

1. *Compute ω in spherical coordinates $(x, y, z) = (\rho \sin\phi \cos\theta, \rho \sin\phi \sin\theta, \rho \cos\phi)$.*
2. *Compute $\mathrm{d}\omega$ in both Cartesian and spherical coordinates and verify that both expressions represent the same 3-form.*
3. *Compute the pullback $\iota^*_{S^2}\omega$ to S^2 using coordinates (ϕ, θ) on the open subset where these coordinates are defined.*
4. *Show that $\iota^*_{S^2}\omega$ is nowhere zero.*

Proof. (1) Using the transformation law

$$\mathrm{d}x^i = \frac{\partial x^i}{\partial \rho}\mathrm{d}\rho + \frac{\partial x^i}{\partial \theta}\mathrm{d}\theta + \frac{\partial x^i}{\partial \phi}\mathrm{d}\phi,$$

we compute

$$\begin{aligned}
\mathrm{d}x &= \sin\phi\cos\theta\mathrm{d}\rho - \rho\sin\phi\sin\theta\mathrm{d}\theta + \rho\cos\phi\cos\theta\mathrm{d}\phi \\
\mathrm{d}y &= \sin\phi\sin\theta\mathrm{d}\rho + \rho\sin\phi\cos\theta\mathrm{d}\theta + \rho\cos\phi\sin\theta\mathrm{d}\phi \\
\mathrm{d}z &= \cos\phi\mathrm{d}\rho - \rho\sin\phi\mathrm{d}\phi.
\end{aligned}$$

We now compute the wedge product using bilinearity and using anticommutativity to ignore terms of the form $\mathrm{d}x^i \wedge \mathrm{d}x^i$:

$$\begin{aligned}
\mathrm{d}y \wedge \mathrm{d}z &= -\sin\phi\sin\theta\mathrm{d}\rho \wedge \rho\sin\phi\mathrm{d}\phi + \rho\sin\phi\cos\theta\mathrm{d}\theta \wedge \cos\phi\mathrm{d}\rho - \rho\sin\phi\cos\theta\mathrm{d}\theta \wedge \rho\sin\phi\mathrm{d}\phi \\
&\quad + \rho\cos\phi\sin\theta\mathrm{d}\phi \wedge \cos\phi\mathrm{d}\rho \\
&= \left(\rho\sin^2\phi\sin\theta + \rho\cos^2\phi\sin\theta\right)\mathrm{d}\phi \wedge \mathrm{d}\rho + \rho\sin\phi\cos\theta\cos\phi\mathrm{d}\theta \wedge \mathrm{d}\rho - \rho^2\sin^2\phi\cos\theta\mathrm{d}\theta \wedge \mathrm{d}\phi \\
&= \rho\sin\theta\mathrm{d}\phi \wedge \mathrm{d}\rho + \rho\sin\phi\cos\theta\cos\phi\mathrm{d}\theta \wedge \mathrm{d}\rho - \rho^2\sin^2\phi\cos\theta\mathrm{d}\theta \wedge \mathrm{d}\phi. \\
\mathrm{d}z \wedge \mathrm{d}x &= -\cos\phi\mathrm{d}\rho \wedge \rho\sin\phi\sin\theta\mathrm{d}\theta + \cos\phi\mathrm{d}\rho \wedge \rho\cos\phi\cos\theta\mathrm{d}\phi - \rho\sin\phi\mathrm{d}\phi \wedge \sin\phi\cos\theta\mathrm{d}\rho \\
&\quad + \rho\sin\phi\mathrm{d}\phi \wedge \rho\sin\phi\sin\theta\mathrm{d}\theta \\
&= -\rho\sin\phi\cos\phi\sin\theta\mathrm{d}\rho \wedge \mathrm{d}\theta + \left(\rho\cos^2\phi\cos\theta + \rho\sin^2\phi\cos\theta\right)\mathrm{d}\rho \wedge \mathrm{d}\phi + \rho^2\sin^2\phi\sin\theta\mathrm{d}\phi \wedge \mathrm{d}\theta \\
&= -\rho\sin\phi\cos\phi\sin\theta\mathrm{d}\rho \wedge \mathrm{d}\theta + \rho\cos\theta\mathrm{d}\rho \wedge \mathrm{d}\phi + \rho^2\sin^2\phi\sin\theta\mathrm{d}\phi \wedge \mathrm{d}\theta. \\
\mathrm{d}x \wedge \mathrm{d}y &= \sin\phi\cos\theta\mathrm{d}\rho \wedge \rho\sin\phi\cos\theta\mathrm{d}\theta + \sin\phi\cos\theta\mathrm{d}\rho \wedge \rho\cos\phi\sin\theta\mathrm{d}\phi - \rho\sin\phi\sin\theta\mathrm{d}\theta \wedge \sin\phi\sin\theta\mathrm{d}\rho \\
&\quad - \rho\sin\phi\sin\theta\mathrm{d}\theta \wedge \rho\cos\phi\sin\theta\mathrm{d}\phi + \rho\cos\phi\cos\theta\mathrm{d}\phi \wedge \sin\phi\sin\theta\mathrm{d}\rho + \rho\cos\phi\cos\theta\mathrm{d}\phi \wedge \rho\sin\phi\cos\theta\mathrm{d}\theta \\
&= \left(\rho\sin^2\phi\cos^2\theta + \rho\sin^2\phi\sin^2\theta\right)\mathrm{d}\rho \wedge \mathrm{d}\phi + \left(\rho\sin\phi\cos\theta\cos\phi\sin\theta - \rho\cos\phi\sin\theta\cos\theta\sin\phi\right)\mathrm{d}\rho \wedge \mathrm{d}\phi \\
&\quad + \left(\rho^2\sin\phi\cos\phi\sin^2\theta + \rho^2\sin\phi\cos\phi\cos^2\theta\right)\mathrm{d}\phi \wedge \mathrm{d}\theta \\
&= \rho\sin^2\phi\mathrm{d}\rho \wedge \mathrm{d}\theta + \rho^2\sin\phi\cos\phi\mathrm{d}\phi \wedge \mathrm{d}\theta.
\end{aligned}$$

Now we assemble the components.

$$\begin{aligned}
\omega &= \rho\sin\phi\cos\theta\left(\rho\sin\theta\mathrm{d}\phi \wedge \mathrm{d}\rho + \rho\sin\phi\cos\theta\cos\phi\mathrm{d}\theta \wedge \mathrm{d}\rho - \rho^2\sin^2\phi\cos\theta\mathrm{d}\theta \wedge \mathrm{d}\phi\right) \\
&\quad + \rho\sin\phi\sin\theta\left(-\rho\sin\phi\cos\phi\sin\theta\mathrm{d}\rho \wedge \mathrm{d}\theta + \rho\cos\theta\mathrm{d}\rho \wedge \mathrm{d}\phi + \rho^2\sin^2\phi\sin\theta\mathrm{d}\phi \wedge \mathrm{d}\theta\right) \\
&\quad + \rho\cos\phi\left(\rho\sin^2\phi\mathrm{d}\rho \wedge \mathrm{d}\theta + \rho^2\sin\phi\cos\phi\mathrm{d}\phi \wedge \mathrm{d}\theta\right)
\end{aligned}$$

$$= \left(-\rho^3 \sin^3\phi \cos^2\theta - \rho^3 \sin^3\phi \sin^2\theta - \rho^3 \sin\phi \cos^2\phi\right) \mathrm{d}\theta \wedge \mathrm{d}\phi = \boxed{\rho^3 \sin\phi \mathrm{d}\phi \wedge \mathrm{d}\theta}.$$

(2) Treat the functions x, y, z as 0-forms. Then given a 2-form η, the wedge product of a 0-form f and η is simply $f \wedge \eta = f\eta$. Applying the product rule for exterior derivatives, $\mathrm{d}(f\alpha) = \mathrm{d}f \wedge \alpha + f \wedge \mathrm{d}\alpha$.

Now apply this to the problem, noting $\mathbb{R}$-linearity and that $\mathrm{d}^2 = 0$ and $\mathrm{d}(x^i) = \mathrm{d}x^i$.

$$\mathrm{d}\omega = \mathrm{d}\left(x\mathrm{d}y \wedge \mathrm{d}z + y\mathrm{d}z \wedge \mathrm{d}x + z\mathrm{d}x \wedge \mathrm{d}y\right) = \mathrm{d}x \wedge \mathrm{d}y \wedge \mathrm{d}z + \mathrm{d}y \wedge \mathrm{d}z \wedge \mathrm{d}x + \mathrm{d}z \wedge \mathrm{d}x \wedge \mathrm{d}y$$

$$= 3\mathrm{d}x \wedge \mathrm{d}y \wedge \mathrm{d}z.$$

In spherical coordinates,

$$\mathrm{d}\omega = \mathrm{d}\left(\rho^3 \sin\phi\right) \wedge \mathrm{d}\phi \wedge \mathrm{d}\theta = \left(\frac{\partial}{\partial\rho}\rho^3 \sin\phi \mathrm{d}\rho + \frac{\partial}{\partial\theta}\rho^3 \sin\phi \mathrm{d}\theta + \frac{\partial}{\partial\phi}\rho^3 \sin\phi \mathrm{d}\phi\right)$$

$$= 3\rho^2 \sin\phi \mathrm{d}\rho \wedge \mathrm{d}\theta \wedge \mathrm{d}\phi + \rho^3 \cos\phi \mathrm{d}\phi \wedge \mathrm{d}\phi \wedge \mathrm{d}\theta = 3\rho^2 \sin\phi \mathrm{d}\rho \wedge \mathrm{d}\theta \wedge \mathrm{d}\phi.$$

To verify that they agree, we compute $\mathrm{d}x \wedge \mathrm{d}y \wedge \mathrm{d}z$ in spherical coordinates:

$$(\sin\phi \cos\theta \mathrm{d}\rho - \rho \sin\phi \sin\theta \mathrm{d}\theta + \rho \cos\phi \cos\theta \mathrm{d}\phi) \wedge (\sin\phi \sin\theta \mathrm{d}\rho + \rho \sin\phi \cos\theta \mathrm{d}\theta + \rho \cos\phi \sin\theta \mathrm{d}\phi)$$

$$\wedge (\cos\phi \mathrm{d}\rho - \rho \sin\phi \mathrm{d}\phi)$$

$$= \text{im not writing this out ok i worked it out on a whiteboard} = \rho^2 \sin\phi \mathrm{d}\phi \wedge \mathrm{d}\theta.$$

Multiplying both of them by 3 shows that they are the same form.

(3) The inclusion $\iota : S^2 \to \mathbb{R}^3$ maps $(\phi, \theta) \mapsto (\rho = 1, \phi, \theta)$. Since d commutes with pullbacks,

$$\iota^*(\omega) = \iota^*(\rho^3 \sin\phi \mathrm{d}\phi \wedge \mathrm{d}\theta) = \sin\phi \mathrm{d}\iota^*\phi \wedge \mathrm{d}\iota^*\theta = \sin\phi \mathrm{d}\phi \wedge \mathrm{d}\theta.$$

Of course we note here that the pullback of a 0-form is simply given by precomposition $\iota^* f = f \circ \iota$, justifying $\iota^*\rho = 1$ and $\iota^*\theta = \theta$, $\iota^*\phi = \phi$.

(4) It's easiest to check this in cartesian coordinates, since these can be defined everywhere. It suffices to show that $(x, y, z) = 0$ never happens, and indeed on the unit 2-sphere we must have $\sqrt{x^2 + y^2 + z^2} = 1$, so not all coordiantes can vanish. Since all the pullback $\iota^*\omega$ does is restrict ω to the sphere, the statement follows. ∎

Problem 14.9

Problem 4.1. *Let M, N be smooth manifolds and $\pi : M \to N$ a surjective submersion with connected fibers. We say $v \in T_pM$ is* vertical *if it vanishes under the pushforward $\pi_{*p}(v) = 0$. Suppose $\omega \in \Omega^k(M)$. Show that there is $\eta \in \Omega^k(N)$ such that $\omega = \pi^*\eta$ if and only if $\iota_v\omega_p = 0$ and $\iota_v \mathrm{d}\omega_p = 0$ for all $p \in M$ and all vertical vectors $v \in T_pM$.*

Recall that

$$\iota_X : \Lambda^k V^* \to \Lambda^{k-1} V^*, \qquad \omega \mapsto \omega(X, -, \cdots, -).$$

Proof. ($\implies$) Let there exist a k-form η with $\omega = \pi^*\eta$. Explicitly, for any $v_1, \ldots, v_k \in T_pM$,

$$\omega_p(v_1, \ldots, v_k) = \eta_{\pi(p)}(\pi_{*p}(v_1), \ldots, \pi_{*p}(v_k)).$$

Let $v \in T_pM$ be vertical. Then

$$\iota_v\omega_p = \omega_p(v, -, \ldots, -) = \eta_{\pi(p)}(\pi_{*p}(v), \pi_{*p}(-), \ldots, \pi_{*p}(-)) = \eta_{\pi(p)}(0, \pi_{*p}(-), \ldots, \pi_{*p}(-)).$$

Recall that the k-covector $\eta_{\pi(p)}$ acts k-multilinearly on k-tuples, and thus we can factor the 0 out:

$$\eta_{\pi(p)}(0, \ldots, \pi_{*p}(-)) = (0\eta_{\pi(p)})(0, \ldots, \pi_{*p}(-)) = 0.$$

By Lee 14.23, $\mathrm{d}0 = 0$ and d commutes with pullbacks, yielding

$$\iota_v(\mathrm{d}\omega_p) = \iota_v(\mathrm{d}\pi^*\eta)_p = \iota_v\pi^*\mathrm{d}\eta_{\pi(p)} = (\mathrm{d}\eta)_{\pi(p)}(\pi_{*p}(v), \pi_{*p}(-), \dots, \pi_{*p}(-)) = (\mathrm{d}\eta)_{\pi(p)}(0, \dots, \pi_{*p}(-)) = 0.$$

($\Longleftarrow$) Let $i_v\omega_p = 0$ and $i_v\mathrm{d}\omega_p = 0$ for all vertical vectors v. For each $q \in N$, choose some $p \in \pi^{-1}(q)$, and for each $u_1, \dots, u_k \in T_qM$ choose a collection $v_1, \dots, v_k \in T_pM$ with $\pi_{*p}(v_i) = u_i$ for all i. Then define $\eta_q(u_1, \dots, u_k) \coloneqq \omega_p(v_1, \dots, v_k)$. Because π is surjective, such a p always exists, and because it is a submersion, it is surjective on each tangent space, meaning such a lift of the vectors always exists. By definition η is alternating and $\pi^*\eta = \omega$, so at this point it suffices only to show that it is well-defined.

First we show independence of lifts $\{v_i\}, \{\widetilde{v}_i\} \subseteq T_pM$ with $\pi_{*p}(v_i) = \pi_{*p}(\widetilde{v}_i)$ for all $1 \le i \le n$. Note that

$$\pi_{*p}(v_i - \widetilde{v}_i) = \pi_{*p}(v_i) - \pi_{*p}(\widetilde{v}_i) = 0,$$

and thus $v_i - \widetilde{v}_i$ is vertical for all i. For convenience of notation, define $w_i \coloneqq v_i - \widetilde{v}_i$. By hypothesis for all $p \in M$,

$$\iota_{w_i}\omega_p = 0, \qquad \iota_{w_i}\mathrm{d}\omega_p = 0.$$

Now consider the difference

$$\omega_p(v_1, \dots, v_k) - \omega_p(\widetilde{v}_1, \dots, \widetilde{v}_k) = \omega_p(v_1, \dots, v_k) - \omega_p(v_1 - w_1, \dots, v_k - w_k).$$

We can expand the second term via multilinearity. Precisely one of these terms must be $\omega_p(v_1, \dots, v_k)$, and all other terms must contain at least one w_i. We then note that

$$\omega_p(v_1, \dots, w_i, \dots, v_k) = (-1)^{i-1}\omega_p(w_i, v_1, \dots, v_k) = (-1)^{i-1}(\iota_{w_i}\omega_p)(v_1, \dots, v_k) = 0$$

since $\iota_{w_i}\omega_p = 0$ by hypothesis. Thus, all terms other than $\omega_p(v_1, \dots, v_k)$ vanish, yielding

$$\omega_p(v_1, \dots, v_k) - \omega_p(\widetilde{v}_1, \dots, \widetilde{v}_k) = \omega_p(v_1, \dots, v_k) - \omega(v_1, \dots, v_k) = 0.$$

Thus ω agrees on lifts.

Now we show η is independent of the choice of p in the fiber of q. It will be important to recall Lee 8.18, which we proved (at least part of, I don't remember if we did the whole thing) in a previous homework.

Proposition 4.1 (Lee 8.18)**.** *Let $F : M \to N$ be a submersion between positive dimensional manifolds M, N. We say $Y \in \mathfrak{X}(M)$ is a* lift *of some $X \in \mathfrak{X}(M)$ if they are F-related. A vector field $V \in \mathfrak{X}(M)$ is vertical if it is F-related to 0. The following are true.*

1. *If $\dim M = \dim N$, then all vector fields on N lift unqiuely.*
2. *If $\dim M \neq \dim N$, then all vector fields lift, but not necessarily uniquely.*
3. *If F is surjective, then each $X \in \mathfrak{X}(M)$ is a lift of a smooth vector field on N if and only if $F_{*p}(X_q) = F_{*q}(X_q)$ for each $F(p) = F(q)$. In this case, it is the lift of a unique vector field.*
4. *If F is surjective and has connected fibers, then $X \in \mathfrak{X}(M)$ is a lift of a smooth vector field on N if and only if $[V, X] = \mathcal{L}_V X$ is vertical whenever $V \in \mathfrak{X}(M)$ is vertical.*

Now we show that η is independent of the choice of $p \in \pi^{-1}(q)$ for $q \in N$. Let $u_1, \dots, u_k \in T_qN$, and for each $p \in \pi^{-1}(q)$, let there exist vector fields V^i for $1 \le i \le k$ with $\pi_{*p}(V^i_p) = u_i$. The existence of these vector fields follows by proposition 4.1 (2), the fact that we may choose a smooth vector field U^i on N with $U^i_p = u_i$, for each i, and the hypotheses of the problem. The problem reduces to showing, for any $p, \widetilde{p} \in M$,

$$\omega_p(V^1_p, \dots, V^k_p) = \omega_{\widetilde{p}}(V^1_{\widetilde{p}}, \dots, V^k_{\widetilde{p}}).$$

Equivalently, we would like to show the function $f : \pi^{-1}(q) \to \mathbb{R}$ defined by

$$p \mapsto \omega_p(V^1_p, \dots, V^k_p)$$

is constant. The function is smooth since evaluation is smooth, and thus it arises as a composite of smooth functions. It suffices to show that its derivative vanishes in any direction. Let V^0 be vertical to π, and by hypothesis

$$0 = \iota_{V^0}\mathrm{d}\omega = \mathrm{d}\omega(V^0, V^1, \ldots, V^k) = V^0(\omega(V^1, \ldots, V^k)) + \sum_{i=1}^{k}(-1)^i V^i(\omega(V, V^1, \ldots, \hat{V}^i, \ldots, V^k))$$

$$+ \sum_{0 \leq i < j \leq k} (-1)^{i+j}\iota([V^i, V^j], V^0, \ldots, \hat{V}^i, \ldots, \hat{V}^j, \ldots, V^k).$$

Note that each term inside the first summation is of the form $(-1)^i V^i(\iota_{V^0}\omega)$, which by hypothesis is zero. Thus the first summation vanishes. For the second term, note that the commutator $[V^0, V^i]$ is vertial for all V since

$$\pi_*[V^0, V^i] = [\pi_* V, \pi_* V^i] = [0, u^i] = 0.$$

We can rewrite terms involving commutators with V^0 as interior products of ω with the commutator, which vanish by hypothesis. If V^0 does not appear in the commutator, then it appears as an argument in ω. Since forms are alternating, we can move V^0 to the first slot, picking up a few factors of -1, and then we obtain an interior product of ω by V, which once again vanishes. Thus all terms in the second summation vanish, yielding

$$V^0(\omega(V^1, \ldots, V^k)) = 0$$

for all vertical V^0. Thus f is constant. ∎

Problem 15.1

Problem 5.1. *Let M be a smooth manifold that is the union of two orientable open submanifolds with connected intersection. Show that M is orientable. Use this to prove S^n is orientable.*

Proof. Choose orientations $\mathcal{O}_U$, $\mathcal{O}_V$ or U and V, respectively. We view orientations as nonvanishing n-forms. Define a function $\sigma : U \cap V \to \{-1, 1\}$ by

$$\sigma(p) = \begin{cases} 1 & \text{if } \mathcal{O}_U|_p = \mathcal{O}_V|_p, \\ -1 & \text{if } \mathcal{O}_U|_p = -\mathcal{O}_V|_p. \end{cases}$$

Since orientations are locally constant (relative to coordinate charts), this function is locally constant, and thus continuous (where $\{-1, 1\}$ has the discrete topology). Because $U \cap V$ is connected, σ must be a constant function. The relevance of this is that if $\sigma(p) = 1$ for all p, then the orientations agree; otherwise we can simply swap $\mathcal{O}_V$ for the orientation $-\mathcal{O}_V$, and the orientations will then agree. Define a global orientation $\mathcal{O} \in \Omega^n(M)$ on M by

$$\mathcal{O}|_p = \begin{cases} \mathcal{O}_U|_p & p \in U \\ \widetilde{\mathcal{O}}_V|_p & p \in V \end{cases}$$

This is well-defined since the orientations agree on the intersection, and by assumption vanishes nowhere and is an n-form. Since open submanifolds have codimension 0, this is a nowhere vanishing n-form on M, and thus an orientation on M.

Let $N, S \in S^n$ be the north and south poles. Let $U = S^n \setminus \{N\}$ and $V = S^n \setminus \{S\}$. Both U and V are diffeomorphic to $\mathbb{R}^n$ by stereographic projection. Since $\mathbb{R}^n$ is orientable, U and V inherit canonical orientations by pulling back along the diffeomorphism. Since they are open submanifolds of S^n, their union is S^n, and their intersection $S^n \setminus \{N, S\}$ is clearly connected, S^n is orientable. ∎